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Centimeter Wave: Next Paradigm for Wireless Communication and Sensing

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FR1 (sub-6GHz) FR3 (cmWave) FR2 (mmWave)
0.41 7.125 24.25 71.0 f (GHz)

Abstract

Centimeter wave (cmWave) band, spanning from 7 GHz to 14 GHz, bridges the gap between sub-6GHz and millimeter wave (mmWave) spectrums, offering an alternative solution for future wireless communication and sensing. CmWave spectrum strikes a balance between the size of antenna array and rich scattering propagation conditions, which facilitates multi-stream multiple-input-multiple-output (MIMO) communications. Additionally, the centimeter-level wavelength enables long-range MIMO radar with superior robustness to undesireable obstacles along the transmission path. Therefore, the potential of cmWave has attracted much attention from both academia and industry. However, the unique wavelength of cmWave signals brings a propagation environment that differs from existing sub-6GHz and mmWave scenarios, introducing new challenges for cmWave transceiver design. To this end, this article investigates the cmWave band in detail and explores its potential for wireless communication and sensing systems. Firstly, we introduce the fundamental characteristic of cmWave band and compare it with existing sub-6G and mmWave counterparts from three aspects of channel model, circuit design and performance evaluation. Subsequently, we illustrate the application scenarios of cmWave band from ultra-high-rate communication to integrated sensing and communication. Finally, we brief the crucial technologies that need to be addressed, providing valuable guidance for future deployment of this field.

Introduction

cmWave的引入

FR1: 频谱资源有限

FR2: 路损高

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FR3: 覆盖广，容量高

(折衷方案)

As information technologies continue to evolve, emerging use cases such as digital twins, immersive communications and autonomous driving, have necessitated concurrent multi-stream data transmission to facilitate unprecedented transmission capacity and sensing capability [1]. To satisfy the requirements of these applications in 2030 and beyond, it is crucial to optimize the utilization of all available frequency bands according to specific service requirements [2, 3]. The existing spectrum for fifth-generation and beyond (5G/B5G) wireless communications is primarily divided into two frequency ranges, that is, Frequency Range 1 (FR1) also known as sub-6GHz band and Frequency Range 2 (FR2) commonly referred to as millimeter wave (mmWave) band. On one hand, the limited spectrum resources of FR1 make it challenging to provide high-rate and high-resolution services. On the other hand, FR2 could support high-rate communication, which is however limited to local dense environments due to its severe propagation loss [4]. To resolve this contradiction, centimeter wave (cmWave) band, staying as a transition frequency range (7–14 GHz) between FR1 and FR2, has garnered much attention from both academia and industry.

CmWave band, which combines the favorable propagation characteristic from sub-6G band together with abundant bandwidth from mmWave band, provides a sweet spot between coverage and capacity. When considering hundreds of MHz bandwidth allocated for each service, cmWave band is well-suited to support Gbps-level communications and submeter-level sensing applications in cellular wireless networks. However, given the anticipated sharp increase in both connection density and bandwidth demand, the spectrum-efficient multiplexing techniques become especially indispensable at cmWave band, represented by multiple-input-multiple-output (MIMO) technology [5]. Thanks to its centimeter-level wavelength, cmWave MIMO is inherently compatible with actual implementations of large-scale antenna array at both base station (BS) and user equipment (UE), also known as cmWave massive MIMO technology. By transmitting and receiving multiple data streams simultaneously over the same frequency channel, cmMave massive MIMO could offer significant channel capacity gains proportional to the minimum number of either transmitting or receiving antennas. Meanwhile, cmWave massive MIMO enables 3-dimensional target positioning and fine-grained motion detection to facilitate future human-machine interactions.

Despite its significant advantages, the research of cmWave band has received relatively little attention compared to its sub-6G and mmWave counterparts, leading to a gap in both scientific research and practical implementation. It risks the efficient utilization of cmWave band that might

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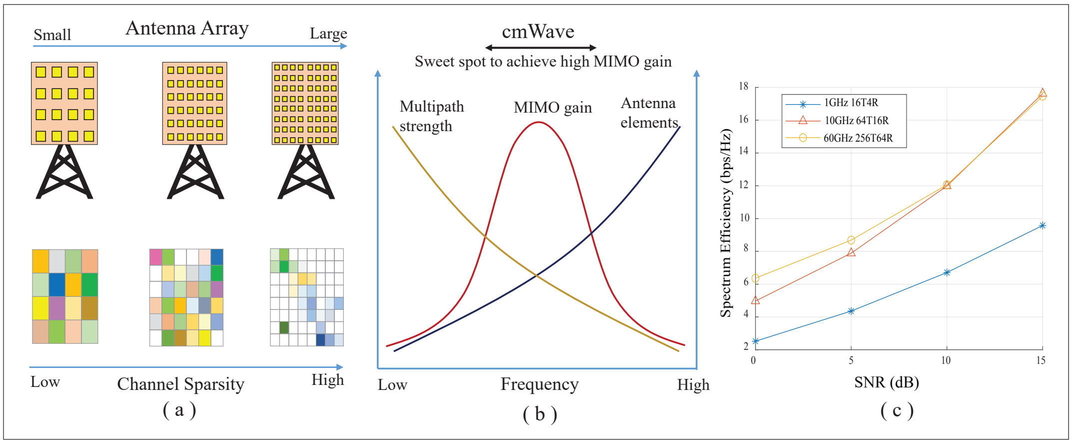

FIGURE 1. Comparison among sub-6GHz, cmWave and mmWave bands with different MIMO configurations, where cmWave band strikes a sweet spot between multipath strengths and antenna elements, hence achieving high MIMO gain.

When signal-to-noise ratio (SNR) is kept unchanged, the more evenly distributed the strength of each multipath component, the greater the MIMO channel capacity.

play a pivotal role in enabling the seamless convergence of communication/sensing in future wireless networks. In this article, we aim to highlight the unique advantages and challenges, therefore fostering the innovation and accelerating the exploration of cmWave band as a key enabler for next-generation wireless systems.

This article presents an extensive survery of current technologies/developments in the field of cmWave band, and systematically analyzes its intrinsic strengths when compared to its existing sub-6GHz band and mmWave counterparts. Subsequently, from the perspective of channel model, circuit design and massive MIMO processing, we detailed the key technologies of cmWave massive MIMO and demonstrate its potential application scenarios. Finally, several challenges and opportunities are highlighted for future research, including cooperative sensing and communication, hardware breakthrough, dynamic spectrum management and low-overhead channel estimation, etc.

CmWave Characteristics

In this section, we introduce cmWave channel, cmWave components and its peformance comparison with existing spectrums.

CmWave Channel

CmWave spectrum exhibits significant multipath effects, creating favorable propagation conditions that make it particularly well-suited for massive MIMO technology. Firstly, the wavelength of cmWave band is smaller than the roughness of common reflective surfaces such as roads, walls, and rooftops, leading to rich scattering phenomena. Additionally, the wavelength of cmWave band is comparable to the size of typical obstacles like pedestrians, trees, and indoor furnitures, enabling efficient diffraction around these objects. Therefore, cmWave band obtains better coverage and multipath gain than its mmWave counterpart, which provides higher data rate and robust tracking ability.

Fig 2

With proper precoding for massive MIMO, different data streams could be transmitted simultaneously over independent multipath channels. When signal-to-noise ratio (SNR) is kept unchanged, the more evenly distributed the strength of each multipath component, the greater the MIMO channel capacity. Since evenly distributed multipath components allow each data stream to fully utilize the spatial diversity, maximizing the available degrees of freedom in the communication channel, and thereby enhancing spectral efficiency and overall data throughput.

As illustrated in Fig. 1, in the sub-6GHz band, the actual deployment of large-scale antenna arrays is constrained by the physical size of the antenna panels, limiting the total channel capacity to the minimum number of either transmitting or receiving antennas. In contrast, severe reflection/diffraction losses in mmWave band exhibit significant channel sparsity, and the channel capacity is primarily influenced by the strength of the multipath components. Both sub-6GHz and mmWave bands cannot fully exploit the potential of massive MIMO, while cmWave band achieves a good balance between the number of independent paths and the strength of multipath components, allowing for the optimal multi-stream transmission gain.

CmWave Components

CmWave band offers advantages in terms of circuit design and device support, particularly in deploying large-scale MIMO technology. For example, at a carrier frequency of 10 GHz, where its wavelength is 3 cm, one smartphone with the size of 14 cm × 7 cm can effectively integrate an 8×4 half-wavelength spacing uniform planar array (UPA). With 32 antenna elements, UEs could work effectively with their serving BS equipped with up to 1024 antennas, enabling a practical massive 1024×32 MIMO system.

Additionally, cmWave band supports fully digital precoding architecture, which provides a distinct advantage over its mmWave counterpart, usually adopting hybrid analog-digital beamforming due to hardware circuit limitations. While mmWave circuits are constrained by both costs and power consumptions associated with digital-to-analog converters and phase shifters, resulting in the reduced flexibility and effectiveness, cmWave band allows for complete digital control over each antenna element [6, 7]. It could significantly enhance the beamforming and signal processing, enabling fine spatial multiplexing and improved user targeting. Consequently, fully digital precoding in cmWave systems could maximize MIMO precoding gains, reduce inter-user interference, and ultimately deliver better communication quality. Overall, the inte-

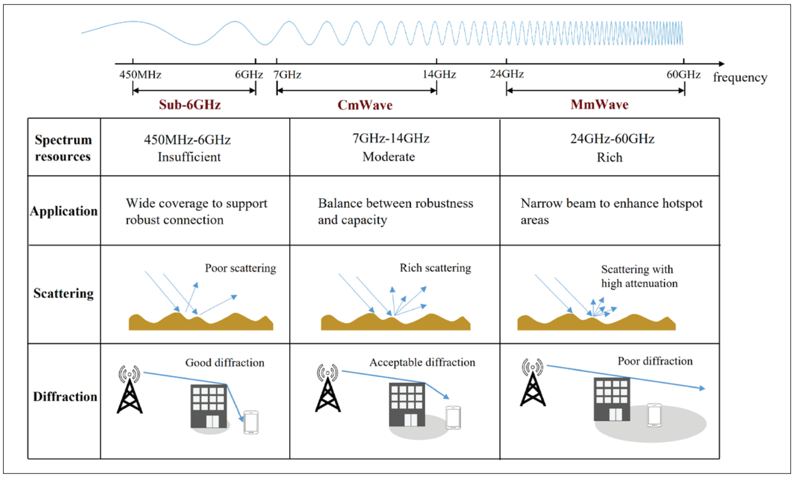

FIGURE 2. Propagation characteristics of sub-6GHz, cmWave and mmWave bands, where cmWave band achieves rich scattering and acceptable coverage to facilitate robust and high-rate communications.

gration of cmWave technology provides a good balance between performance and practical implementation, which might address the challenges faced by mmWave band, making it a compelling candidate for future wireless systems.

Comparison with Existing Spectrums

Current frequency bands used for 5G/B5G wireless communications primarily consist of sub-6GHz and mmWave bands. As shown in Fig. 2, the main advantage of sub-6GHz band lies in its longer propagation distance, which effectively supports wide-area coverage. Due to its relatively low carrier frequency, sub-6GHz signals exhibit strong penetration/diffraction capabilities, enabling better performance in overcoming the negative effects from obstacles such as buildings and ensuring stable signal transmission. Additionally, based on the previous advancements made in sub-3GHz technologies, sub-6GHz band benefits from high technological maturity with well-established devices and infrastructures, which could reduce the complexity/cost of actual deployments and maintenances. However, the narrow bandwidth of sub-6GHz spectrum limits its ability to support ultra-high-rate transmission. Furthermore, although sub-6GHz band satisfies favorable propagation condition, its long wavelength and limited number of antennas restrict the adoption of massive MIMO technology, particularly at the UE side, where the actual deployment of large-scale antenna array becomes challenging.

In contrast, mmWave band offers substantial available bandwidth, providing significant enhancements in capacity compared to its sub-6GHz counterpart. However, high-frequency electromagnetic waves in the mmWave band are subject to considerable path loss, poor penetration through obstacles, and are highly susceptible to blockage from surrounding objects. These limitations make it challenging to realize comprehensive wide-area coverage. Consequently, the primary application of mmWave band concentrates on the deployment of massive antenna array in specific hotspot areas, such as dense urban environments or high-traffic locations, where the demand for ultra-high-rate services is critical.

CmWave band, as a bridge between sub-6GHz and mmWave bands, achieves a good balance between robustness and capacity. By combining the mature sub-6GHz technology with the emerging millimeter wave technology, cmWave band could exploit its advantages and circumvent its weaknesses. Due to its moderate wavelength, cmWave signals can diffract around various small obstacles, which maintain stable communication in dynamics wireless conditions, supporting high data rates while avoiding frequent call drops. On the other hand, cmWave band leverages a wider bandwidth compared to its sub-6GHz counterpart. The integration of massive MIMO technology enables spatial multiplexing, greatly enhancing the spectrum efficiency and accommodating a larger number of connected devices. Furthermore, cmWave band is capable of meeting the precision requirements for integrated sensing and communication (ISAC), which offers significant advantages for actual network deployment and service expansion in real-world scenarios.

CmWave Application Scenarios

In this section, we illustrate the potential application scenarios of cmWave band.

Ultra-High-Rate Communication

The performance metrics of 5G/B5G wireless systems are entirely advanced compared to previous generations of wireless communications, with key indicators designed to support diverse applications, including enhanced mobile broadband (eMBB), ultra-reliable low-latency communication (URLLC), and massive machine-type communication (mMTC). With the widespread deployment of 5G/B5G networks, users can experience downlink speeds of 100 Mbps in everyday scenarios, and can even reach several Gbps in hotspot areas, which have significantly enhanced wireless communication capabilities, enabling support for applications such as high-definition video and high-rate live streaming. These advancements have revolutionized entertainment experiences, allowing users to enjoy seamless, real-time interactions with improved visual and audio qualities.

It is expected that 6G wireless communications not only address the legacy issues of 5G/B5G wireless systems but also expand the application scenarios. Immersive entertainment demands ultra-high-rate connections with robust posture stability, while smart homes, smart traffic, and smart cities impose strict requirements for coverage capability and low latency in dynamic wireless environment. To meet the aforementioned service demands, robust connectivity is required in sub-6GHz bands as well as ultra-high-data transmission capability in mmWave band. As a result, many researches focused on achieving the cooperative integration of sub-6G/mmWave bands through efficient switching or dual connectivity [8]. It is evident that cmWave band could harness the advantages of both sub-6G and mmWave bands. Under favorable conditions, cmWave technology might support about 20 data streams for each UE, enabling ultra-high-rate communications, which is considered as a promising candidate for 6G wireless communications.

Integrated Sensing and Communication

CmWave band exhibits distinct advantages for ISAC applications, particularly due to its wavelength alignment with the characteristics of physical targets [9]. Unlike mmWave signals whose images are close to optical images, cmWave signals can diffract around obstacles like trees, pedestrians, and furniture, which allows to capture critical information about the target's position, velocity, and posture changes for ISAC wireless systems, providing a solid technical foundation for immersive applications and digital twins. Figure 3 presents an example of cmWave ISAC system, which can effectively mitigate the effects of structural occlusions and guarantee the sensing capabilities from higher frequency bands, thereby improving the overall point cloud reconstruction quality.

Moreover, cmWave band could be seamlessly integrated with existing C-band/X-band services, such as weather radar, aeronautical communications, etc. The integration of large-scale antenna array in cmWave band, together with fully digital signal processing, could enhance the angular resolution significantly. The feasibility to deploy a substantial number of antennas allows for finer spatial sampling and improved resolution between closely spaced targets. Leveraging the fully digital precoding/decoding architecture of cmWave band, angular/range super-resolution sensing could be realized through advanced optimization algorithms [10], enabling applications such as autonomous vehicles, smart traffic scheduling, and high-precision robotic control, etc. The synergy between robust communication links and accurate sensing capabilities underscores the potential of cmWave band, making it an essential candidate for the evolution of future wireless networks.

CmWave Key Technologies

Despite the intrinsic advantages of cmWave band, it is challenging to fully unleash massive cmWave MIMO gains. For example, the performance degradation of precoding in hybrid near-field and far-field scenarios, the actual deployment of large-scale antenna arrays, and the high computational complexity associated with fully digital beamforming need to be carefully investigated. To overcome these limitations, several key technologies are crucial and need to be carefully investigated in practical scenarios, which are illustrated in Fig. 4.

Unified Far-Field and Near-Field

Traditional sub-6GHz channel model usually has a far-field assumption, where electromagnetic waves propagate as plane waves. Conversely, in mmWave systems employing large antenna array, many studies adopt a purely near-field model based on spherical wave propagation. As illustrated in Fig. 5, when considering cmWave band and large antenna array, Rayleigh distance typically extends to the scale of hundreds of meters. In a typical cmWave cell with a radius of several hundred meters, the coexistence of both near-field and far-field scatterers leads to a complex hybrid-field superposition between BS and UEs, which significantly complicates channel modeling, channel estimation, and precoding [11]. Consequently, it is crucial to develop a unified model which can seamlessly integrate near-field/far-field characteristics. An effective unified far-field and near-field channel model could facilitate accurate channel representation, optimize beamforming strategy, and enhance the performance of cmWave massive MIMO systems.

One promising approach to bridge the far-field/near-field gap is to adopt the wavenumber

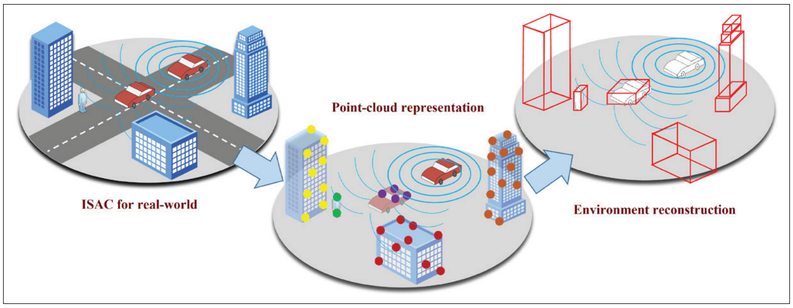

FIGURE 3. CmWave ISAC could accurately reconstruct the wireless environment with moderate wavelength and bandwidth.

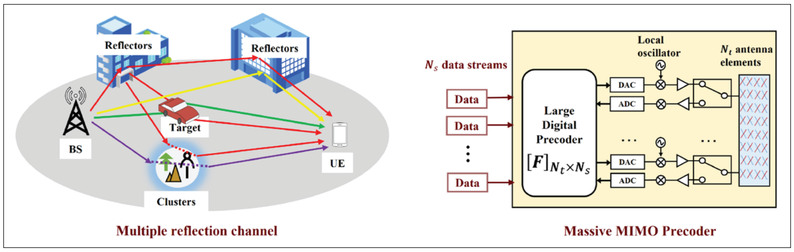

FIGURE 4. CmWave band introduces significant challenges for channel estimation and MIMO precoding.

domain representation to unify both far-field plane waves and near-field spherical waves. By capturing the essential characteristics of hybrid-field propagation, it may lead to more efficient channel estimation and beamforming algorithms, paving the way to overcome the degradation introduced by hybrid-field effect. Also, AI-based technology has the potential to capture the high-dimensional characteristics, which may create a digital twin environment to obtain the complex features. The ability to efficiently estimate the unified far-field/near-field channels and apply tailored precoding strategy under hybrid-field conditions is critical to realize the full potential of cmWave band in 6G wireless networks.

Antenna Architecture

To unlock the potential of cmWave band for 6G communication and sensing, technical advancements in antenna architectures are indispensable. At cmWave band, the moderate wavelength allows for the deployment of larger antenna arrays when considering practical physical dimensions, which is essential for achieving high spectral efficiency and enhanced sensing capabilities simultaneously. Due to the constraint of actual deployment costs, it is important to develop flexible and cost-effective massive antenna array architectures to improve the size of antenna array, adaptability, performance, etc.

Reconfigurable intelligent surface (RIS) has drawn much attention due to its inherent ability to reconfigure the wireless propagation environment. By strategically placing RIS panels composed of numerous passive reflecting elements, the spatial distribution of cmWave signals can be dynamically adjusted to enhance the coverage, improve the SNR, and reduce the interference. One advanced RIS, named as holographic RIS, employs densely packed and ultra-thin electromagnetic elements to achieve continuous and precise wavefront shaping, which creates holographic beam patterns, significantly improving beamforming accuracy and spatial resolution. For

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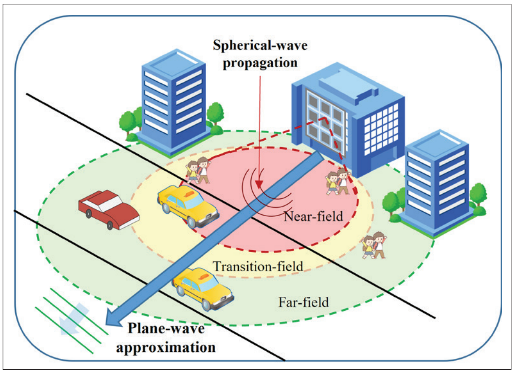

FIGURE 5. Different signal propagation models for far-field and near-field, where plane-wave assumption is applied beyond Rayleigh distance and spherical-wave effect must be considered within Rayleigh distance. The yellow area near Rayleigh distance is called transition field.

cmWave band, holographic RIS could enhance both communication and sensing capabilities, offering unprecedented control over signal propagation and environmental perception.

Flexible antenna panels leverage materials like stretchable conductive polymers and nanomaterials to generate conformable antenna arrays. These panels can adapt to non-flat surfaces, such as the curved fuselage of drones or wearable devices, making them ideal for diverse deployment scenarios. Flexible antenna arrays also enable the seamless integration of cmWave technology into emerging applications such as smart textiles and internet of things (IoT) devices. Irregular antenna arrays, characterized by non-uniform spacing and configurations, could reduce grating lobes and enhance spatial resolution. These arrays are particularly suited for multipath environment, leveraging the favorable propagation conditions of cmWave massive MIMO.

In brief, advanced antenna architectures, such as RIS, holographic RIS, flexible panels and irregular antenna arrays are cornerstones of cmWave band. By deployment of larger, more adaptive antenna arrays, the unique challenges of cmWave could be solved, which paves the way for high-capacity, high-precision 6G ISAC systems.

Fully Digital Beamforming

In cmWave band, another critical challenge lies in high-dimensional channel estimation [12]. The adoption of cmWave massive MIMO significantly increases the size of the channel matrix, leading to huge pilot overhead to acquire full channel state information (CSI). Current mmWave systems adopt codebook-based channel estimation strategies (such as Type I and Type II codebooks) in order to reduce the overhead of channel estimation. By leveraging the sparsity of mmWave channels in the angle-delay-Doppler domain, channel estimation is performed in the sparse transform domain, and only the parameters related to significant paths are fed back. In cmWave systems, unfortunately this sparsity-based approach becomes less effective. Unlike mmWave systems, where wireless channels exhibit strong sparsity, cmWave propagation is characterized by richer multipath conditions due to its favorable diffraction/scattering properties, where the loss of structural sparsity makes it difficult to identify a sparse representation in the traditional Fourier domain. Consequently, codebook-based methods, which rely on sparsity, fail to adequately represent the channel in cmWave systems, leading to higher channel estimation errors. Given that the performance of cmWave massive MIMO is highly dependent on accurate channel estimation, it is important to investigate high-accuracy and low-overhead channel estimation techniques.

A promising trend is to develop more accurate channel models for cmWave band, which focuses on extracting key characteristics of cmWave channel and using more effective parameters to capture its essential features. For instance, cmWave channel could be decomposed into long-term statistical channel information and instantaneous channel variations, which could reduce the overhead of channel estimation by decreasing the frequency of channel measurements. Alternatively, employing CSI compression techniques, such as appropriate auto-encoders/decoders by neural network, could significantly compress the CSI before feedback, thereby reducing the communication overhead [13]. Furthermore, large-scale digital twin technology may be applied for channel acquisition. By pre-establishing digital twin model, BS could potentially retrieve channel information, or fingerprints, for any given location within its serving cell, which allows for real-time channel estimation without extensive pilot transmissions, offering a significant overhead reduction while maintaining accurate CSI.

After effectively estimating cmWave channel, traditional precoding and decoding methods face exponentially growing computational complexity as the number of antennas increases, which is very challenging for real-time applications requiring low-latency processing [14]. As mentioned previously, traditional Type I/II codebook-based precoding methods suffer high performance degradation due to less structured cmWave channels under favorable propagation conditions. Therefore, it is essential to design precoding/decoding schemes specifically tailored for cmWave systems.

One possible approach is to remodel the cmWave channel and think about a more suitable transform domain to construct the precoding/decoding operators. By optimizing these operators, it is possible to reduce the computational complexity in an iterative manner. Alternatively, a layered precoding strategy can be employed, where users are grouped to reduce the optimization space, thus achieving a balance between complexity and performance. These techniques could mitigate the problems posed by the complex structure of cmWave channels while maintaining the efficient signal processing and communication quality.

Open Issues and Future Prospects

In this section, we highlight several open problems of cmWave communication and sensing together with the corresponding research opportunities, from the perspective of hardware implementation, spectrum management and signal processing strategies.

Hardware Breakthrough

The employment of massive MIMO technologies can be mandatory to fully exploit the potential of cmWave spectrum, which, however, causes additional difficulties in hardware implementation.

Specifically, unlike the compact design of phased-array antenna at mmWave frequencies, traditional antenna array for cmWave massive MIMO can be extremely bulky attributed to its centimeter-level wavelength, which are especially undesirable for non-terrestrial platform stations (e.g., unmanned aerial vehicles) with limited payload carrying capability. One potential solution is the distributed MIMO that combines multiple small arrays into an equivalent large-scale one, thereby expanding the antenna aperture. Besides, the emerging holographic MIMO philosophy and flexible antennas leverage continuous aperture technology to surmount the constraint of half-wavelength antenna spacing, and joint phase time arrays provides delay element to resist the beam squint. On the other hand, to address the power consumption issue associated with fully digital architecture, the development of energy-efficient components becomes a critical focus in cmWave massive MIMO, e.g., low-power radio-frequency (RF) components and low-loss dielectric materials. Another promising solution is the deployment of reconfigurable metasurface array at the transceiver, which establishes an RF-chain-free architecture with competitive beamforming capability. Further breakthrough will surely facilitate the practical application of cmWave massive MIMO technologies.

Dynamic Spectrum Management

For a plethora of mobile services under congestion of the commercial sub-6GHz/mmWave bands, the cmWave spectrum can be incorporated to complement the existing radio resources, which necessitates dynamic spectrum sharing techniques. The emerging intelligent spectrum allocation strategies, leveraging machine learning or even large language models for real-time adaptation, are expected to realize efficient coexistence and cooperation (e.g., carrier aggregation) with existing communication systems. These AI-driven approaches offer advantages such as predictive modeling of traffic patterns, environment-aware spectrum sensing, and real-time decision-making for the optimal resource allocation. However, they also face challenges including the need for large, high-quality datasets, model interpretability, and computational overhead in deployment. Furthermore, cognitive radio techniques can be indispensable for seamless integration of the cmWave communication network into existing counterparts while optimizing the spectrum efficiency.

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Low-Overhead Channel Feedback

Accurate channel estimation is one prerequisite for effective MIMO precoding, which inevitably introduces severe computational and feedback overhead of the channel state information (CSI) especially for cmWave massive MIMO equipped with fully-digital large-scale antenna array. These motivate continued researches on classical channel feedback techniques, such as compressed sensing and codebook design, and next-generation AI-enabled strategies, including joint source-channel coding and semantic communications. Additionally, the channel estimation, CSI feedback and even the precoding modules can be jointly designed with adaptive optimization or even replaced by edge-to-edge neural network architecture, which can enhance both the communication performance and compression ratio of CSI feedback concurrently.

Cooperative Sensing and Communication

As highlighted above, cmWave massive MIMO systems are particularly well-suited to address the high-precision requirement of next-generation sensing/positioning applications attributed to desirable aperture in both frequency and spatial domains. Independent deployment of cmWave MIMO communication and radar subsystems is impractical for structure engineering due to bulky size of the antenna array, not to mention the nearly double bandwidth, energy and hardware expenses. Therefore, the fusion of communication and sensing modules regarding transceiver structure, functions and even transmit waveform, i.e., the ISAC philosophy, should be carefully investigated for cmWave spectrum, which is still at its infancy [15]. For instance, cmWave massive MIMO systems tend to employ the fully-digital architecture, where classical ISAC beamforming technologies cannot apply due to considerable computational overhead. This necessitates innovative low-complexity beamforming strategies for real-time balance between accurate object tracking and superior throughput. Additionally, the convergence of communication, sensing, and computing through shared wireless and hardware resources can introduce sophisticated mutual interference between various functions, which requires effective interference management strategies, particularly within edge-intelligent networks.

Conclusion

In this article, we provided a comprehensive survey on cmWave wireless communications. Compared to its existing sub-6GHz and mmWave bands, cmWave is more suitable for actual implementation massive MIMO wireless systems. We illustrated the advantages of cmWave band by exploring its favorable propagation conditions, strong component support, and superior performance in typical ISAC applications. Finally, we illustrated the challenges that must be addressed for the practical deployment of cmWave band, providing guidance for future research in this field.

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